

NIKE SALES ANALYTICS PROJECT REPORT

1. Project Title

Nike India Sales Data — Cleaning & Analysis

2. Project Overview

This project is about analyzing Nike sales data using Python and Power BI. First, the raw dataset was cleaned in Google Colab by fixing missing values, removing errors, and preparing the data for analysis. After cleaning, the data was imported into Power BI to create an interactive dashboard.

The dashboard helps us understand sales performance by showing revenue, profit, units sold, sales channels, product lines, and regional sales. It makes it easier to identify trends and supports better business decisions.

3. Problem Statement

The Nike sales dataset contained raw data that needed cleaning before analysis. The goal of this project is to clean the dataset, analyze the sales information, and create a Power BI dashboard that provides useful insights into revenue, profit, customer preferences, and regional performance.

4. Objectives

- Clean the raw Nike sales data using Python.
- Handle missing values and remove duplicate records.
- Prepare the dataset for analysis.
- Build an interactive Power BI dashboard.
- Analyze revenue, profit, and units sold.
- Compare sales across different regions and sales channels.
- Understand customer preferences by gender, size, and product line.

- Provide useful business insights and recommendations.
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5. Data Source

- Source: **Kaggle Open Data Portal**
 - Dataset Size: **2500rows × 13 columns.**
 - Timeline: **Data collected during 2024 to 2025.**
 - Domain: **Retail & Sports Industry**
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6. Tools and Technologies Used

- Programming Language: **Python.**
 - Libraries: **Pandas, NumPy, Matplotlib, Seaborn.**
 - Dashboard Creation : **Microsoft Power BI Desktop**
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7. Tools and Technologies Used

Column	Data Type	Description
Order_ID	int64	Unique identifier for every order
Gender_Category	object	Gender of the customer
Product_Line	object	Category of product
Product_Name	object	The name of the product

Size	object	The size of the product that customer purchased
Units_Sold	int64	How many product sold
MRP	float64	Price of the product
Discount_Applied	float64	Percentage of discount
Revenue	float64	Total amount of sales
Order_Date	object	Date of the product purchased
Sales_Channel	object	Mode of sales (Online or Offline)
Region	object	Country of the customer
Profit	float64	Profit made by the sales
order_month	object	Month of the order purchased

8. Data Preprocessing

Before creating the dashboard, the dataset was cleaned using Python.

The preprocessing steps included:

- Handling missing values.
- Checking and removing duplicate records.
- Correcting data types where required.
- Standardizing column values.
- Preparing the cleaned dataset for Power BI visualization.

These steps improved the quality and reliability of the analysis.

9. Dashboard Description

The Power BI dashboard contains the following visualizations:

- **Revenue by Region**

Shows how much revenue was generated in each region.

- **Revenue by Region and Sales Channel**

Compares revenue from Online and Retail sales across regions.

- **Units Sold by Gender Category**

Displays units sold for Men, Women, and Kids products.

- **Revenue and Profit by Order Month**

Shows monthly changes in revenue and profit.

- **Units Sold by Size and Gender Category**

Compares units sold for different sizes and gender categories.

- **Product Line**

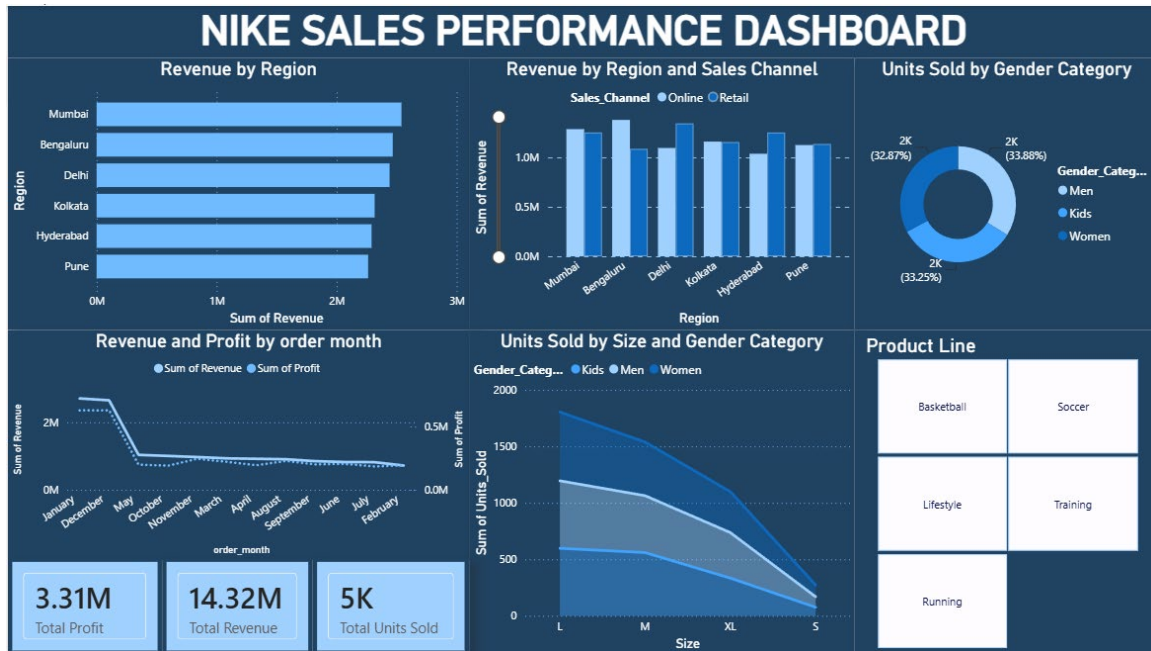
Displays sales across Basketball, Soccer, Running, Lifestyle, and Training product lines.

- **KPI Cards**

The dashboard highlights three important metrics:

- Total Revenue: 14.32 Million
- Total Profit: 3.31 Million
- Total Units Sold: 5 Thousand

Dashboard is Given below



10. Key Findings

- Mumbai and Bengaluru generated the highest revenue among the regions.
- Both Online and Retail channels contributed significantly to overall sales.
- Men, Women, and Kids categories showed almost equal sales distribution.
- Revenue and profit changed across different months, indicating seasonal trends.
- Large and Medium sizes sold more units than Small sizes.
- All major product lines contributed to total sales, with balanced performance.

11. Recommendations

- Increase stock availability in high-performing regions.
- Improve marketing for regions with lower sales.
- Focus on popular product sizes to meet customer demand.
- Promote products with lower sales through offers and discounts.
- Continue monitoring monthly trends for better planning.

12. Final Insights

The dashboard provides a clear overview of Nike's sales performance. It helps identify top-performing regions, customer preferences, and monthly trends. Cleaning the data before analysis improved accuracy and made the dashboard more reliable for business decisions.

13. Conclusion

This project successfully converted raw Nike sales data into meaningful insights using Python and Power BI. After cleaning and analyzing the data, an interactive dashboard was created to visualize important business metrics. The project demonstrates how data analytics can help organizations understand sales performance and make informed decisions for future growth.